

Predicting new COVID-19 cases in the US: A Comparison of a Compartmental model (SIR) and XGBOOST Machine Learning Model

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Background: Figuring out the number of new COVID-19 cases was key in aligning interventions to hotspots. This included ensuring there were enough beds for admissions, key protection gear like face masks, oxygen supply and intensive care unit space. A lot of mechanistic models were build borrowing from the traditional SIR model. Additionally, researchers explored machine learning techniques to predict the course of the infections.

Methods: In this report, we compare the performance of a SIR model to an XGBOOST prediction model in predicting new cases in each of the American states, 3 years after the pandemic.

Results: The XGBOOST ML approach demonstrated robust goodness-of-fit, effectively capturing the non-linear peak dynamics across varying geographic replicates.

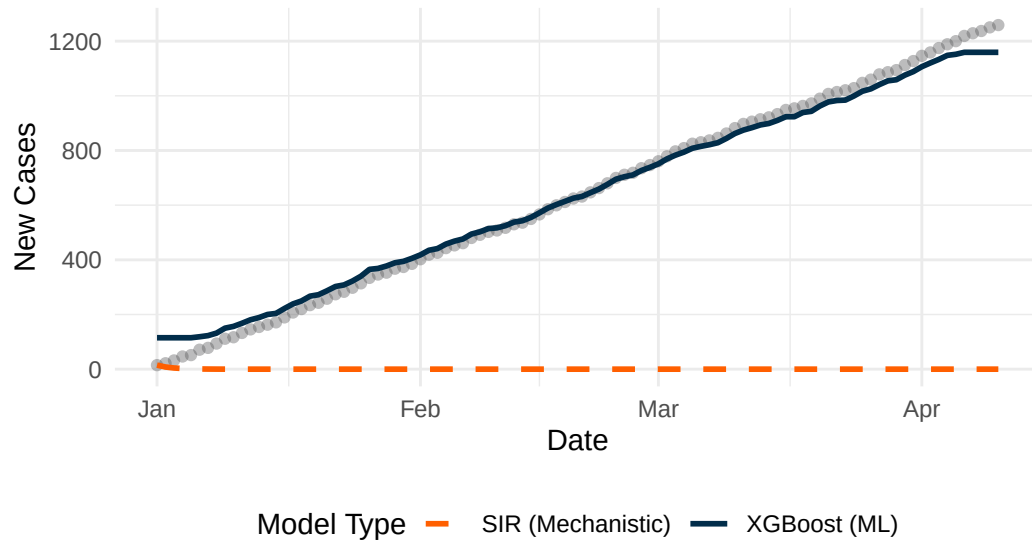
1 INTRODUCTION

We chose to use data on COVID-19 cases assembled by a team from [The London School of Health & Tropical Medicine](#)

Model comparisons

Daily Case Projections

Comparison of ML and SIR outputs



Summary of parameters

Table 1: SIR Model Parameters

Parameter	Value
Beta (Transmission)	0.1869
Gamma (Recovery)	0.8036
Basic Reproduction (R_0)	0.2325

Simulations

Table 2: Global Assessment Metrics

rmse	mape
1.460515	99.80213

XGBoost Fit vs Simulated Data

State 1 – Orange line represents model prediction

